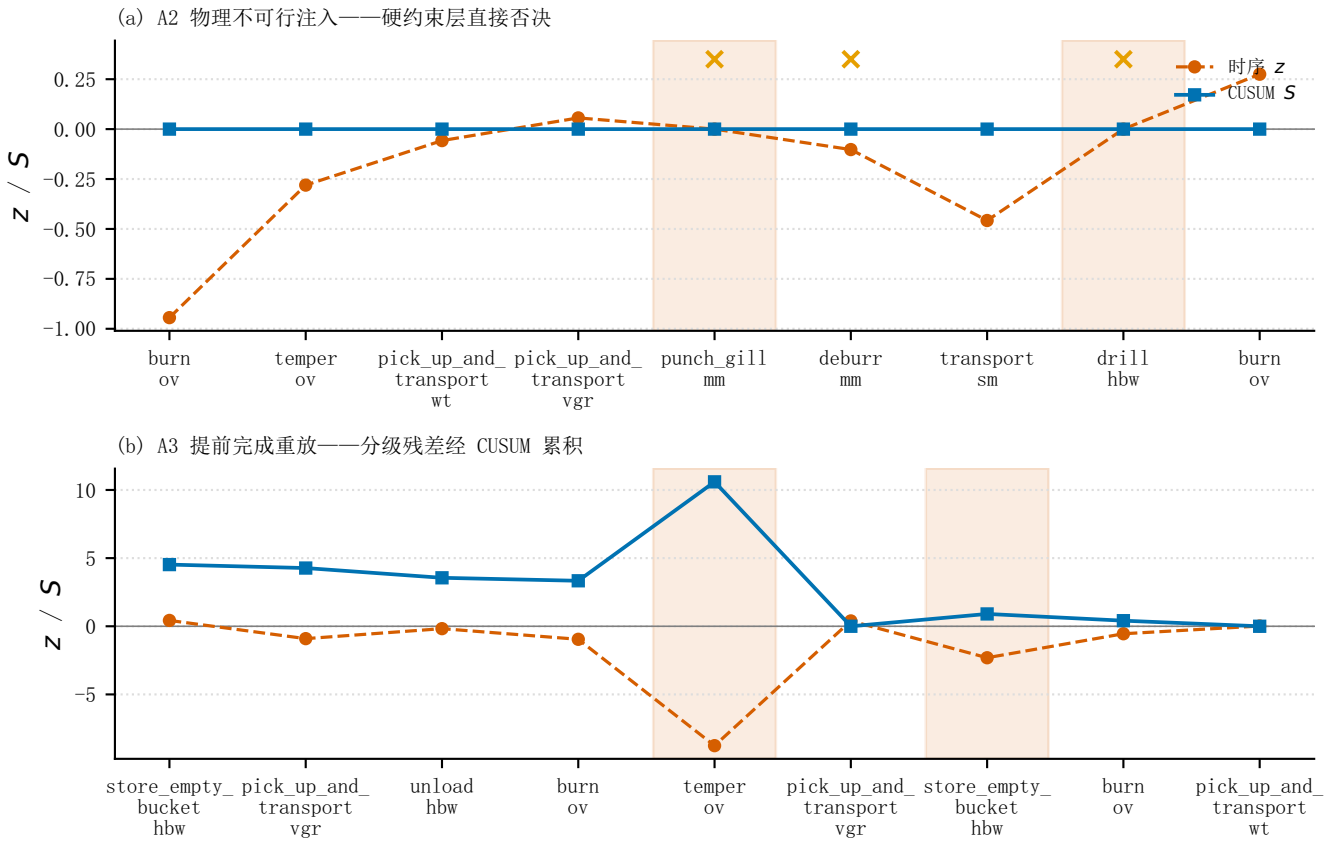
(a) A2 物理不可行注入——硬约束层直接否决
时序 z
CUSUM S
z / S
0.25
0.00
−0.25
−0.50
−0.75
−1.00
burn ov
temper ov
pick_up_and_ transport wt
pick_up_and_ transport vgr
punch_gill mm
deburr mm
transport sm
drill hbw
burn ov
(b) A3 提前完成重放——分级残差经 CUSUM 累积
z / S
10
5
0
−5
store_empty_ bucket hbw
pick_up_and_ transport vgr
unload hbw
burn ov
temper ov
pick_up_and_ transport vgr
store_empty_ bucket hbw
burn ov
pick_up_and_ transport wt